Project Scheduling Techniques

12.1 Introduction

Any project involves *planning*, *scheduling* and *controlling* a number of interrelated activities with use of limited resources namely, *men*, *machines*, *materials*, *money* and *time*. The projects may be extremely large and complex such as construction of a power plant, a highway, a shopping complex, ships and aircraft, introduction of new products and R&D projects. *It is required that managers must have a dynamic planning and scheduling system to produce the best possible results and also to react immediately to the changing conditions and make necessary changes in the plan and schedule*. A convenient analytical and visual technique of ***PERT(Project Evaluation and Review Technique)*** and ***CPM(Critical Path Method)*** prove extremely valuable in assisting the managers in managing the projects.

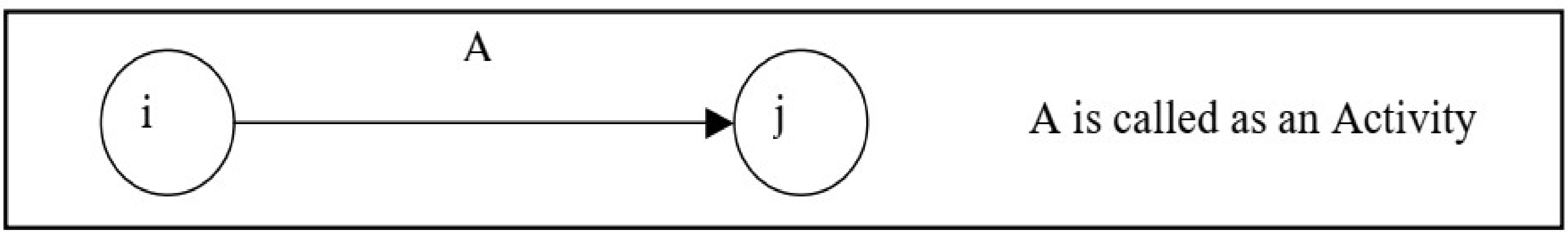
12.2 PERT/CPM Network Components

PERT/CPM networks contain *two major* components

i. Activities

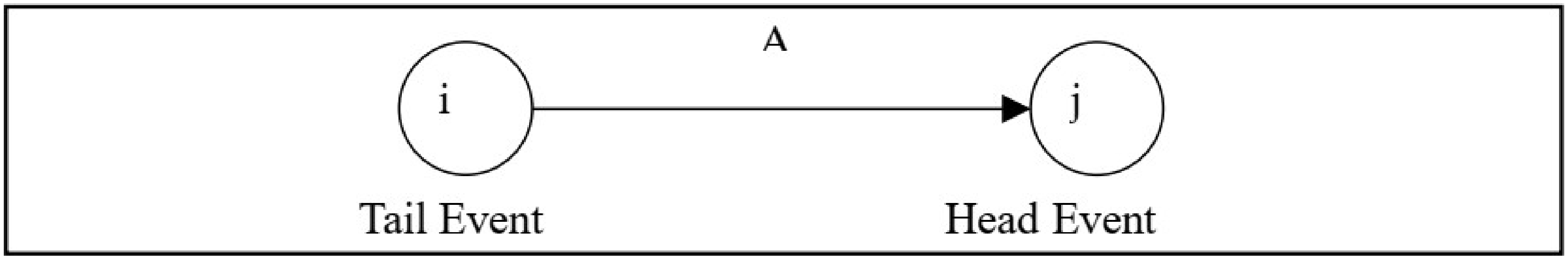
ii. Events.

Activity: An activity represents *an action* and *consumption of resources* (time, money, energy) required to *complete a portion of a project*. *Activity is represented by an arrow* as shown below:

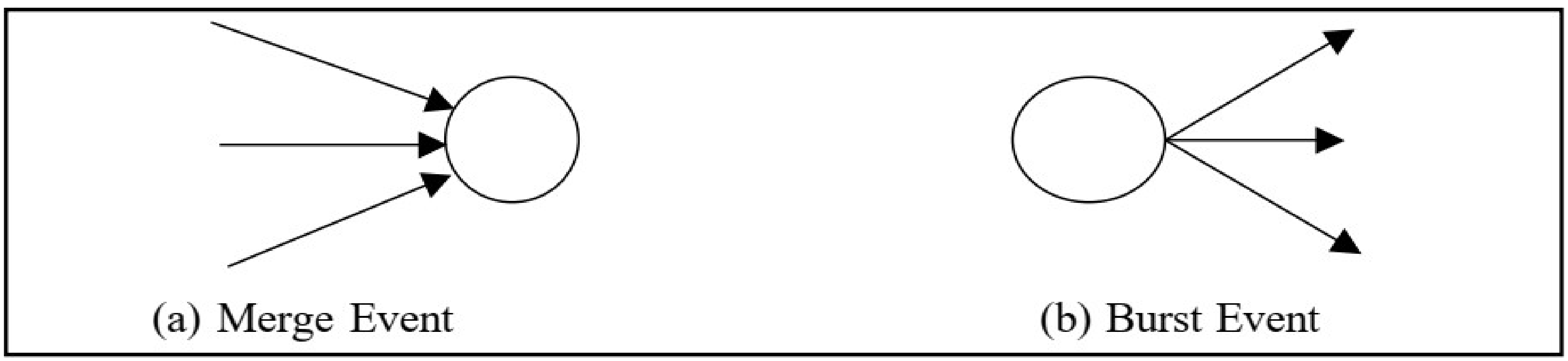


12.2 PERT/CPM Network Components (continued…)

Event: An event(or node) will always *occur at the beginning and end of an activity*. The event has *no resources* and *is represented by a circle*. The i^{th} *event and* j^{th} *events are the tail event and head event* respectively as shown below:

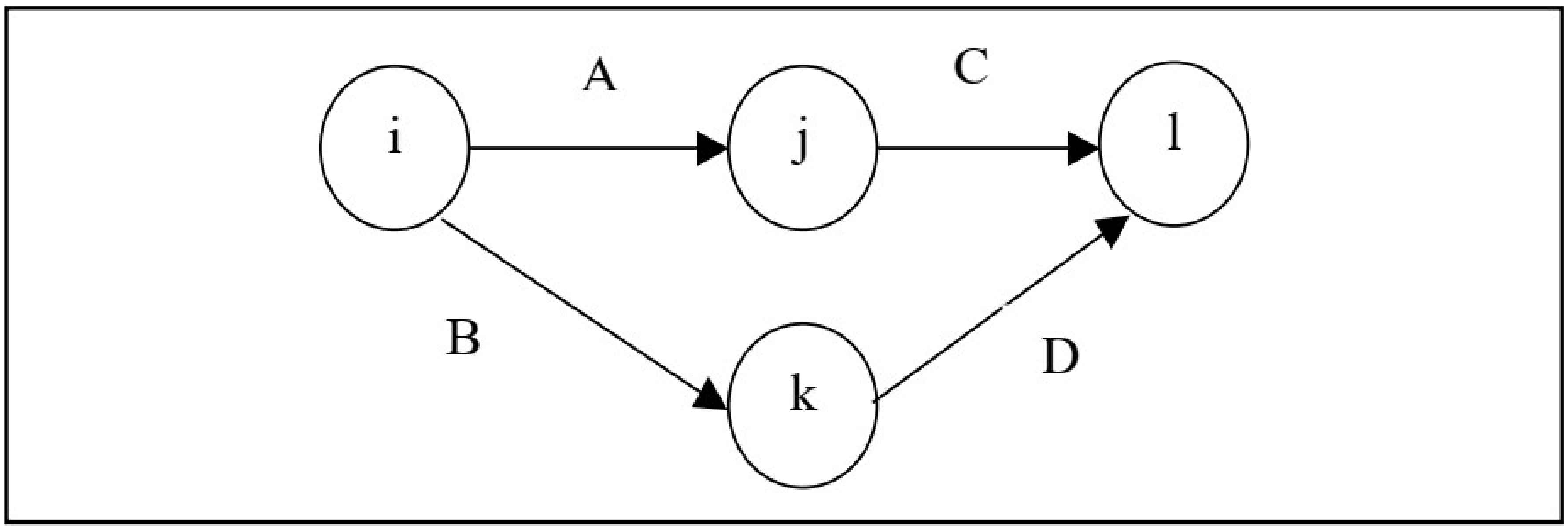


Merge and Burst Events: *One* or *more activities can start and end simultaneously at an event* as shown below:



12.2 PERT/CPM Network Components (continued…)

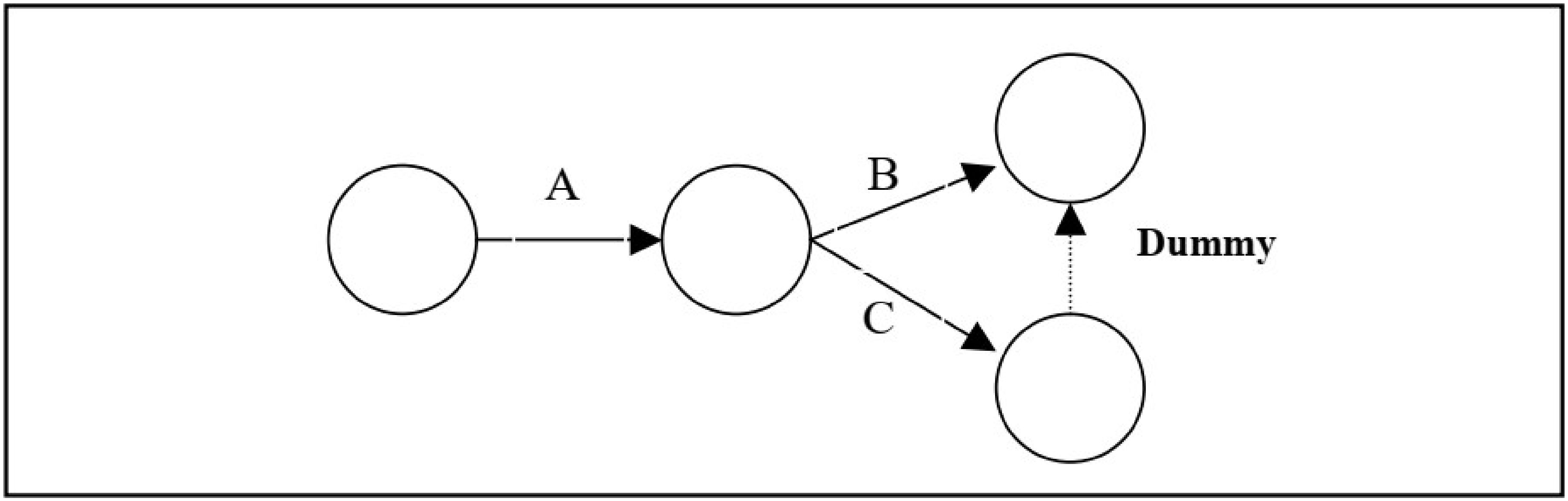
Preceding and Succeeding Activities: *Activities performed before given events* are known as ***preceding activities*** and *activities performed after a given event* are known as ***succeeding activities*** as shown below:



Activities A and B precede activities C and D respectively.

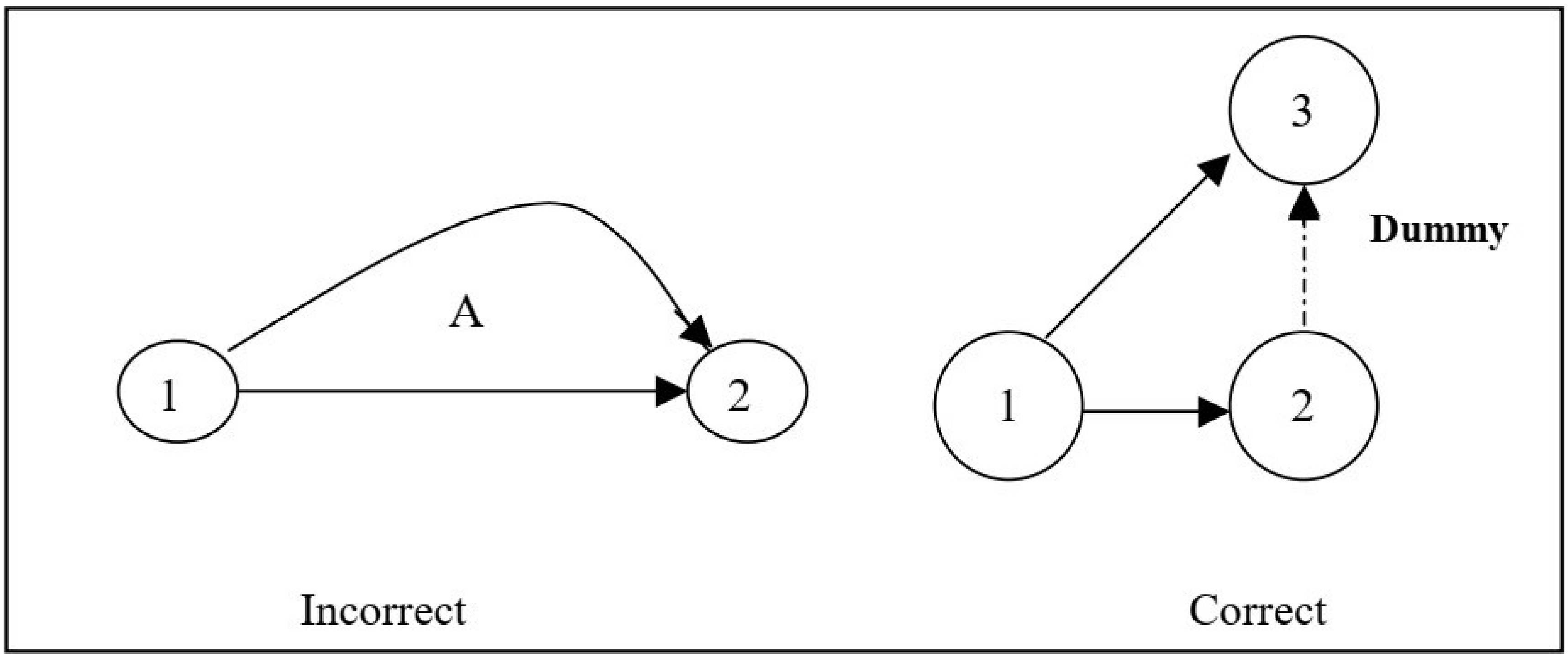
12.2 PERT/CPM Network Components (continued…)

Dummy Activity: An ***imaginary*** *activity which does not consume any resource and time* is called a *dummy activity*. Dummy activities are simply used *to represent a connection between events in order to maintain a logic in the network*. It is represented by a dotted line in a network as shown below:



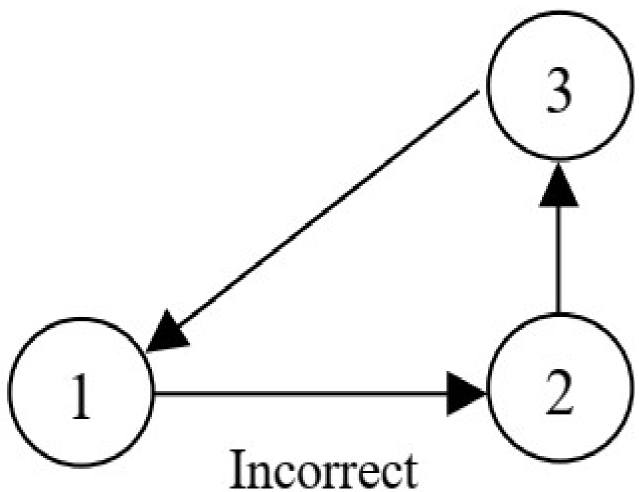
12.3 Errors to be Avoided in Constructing a Network

a. *Two activities starting from a same tail event must not have a same end event.* To ensure this, it is absolutely necessary *to introduce a dummy activity* as shown below:



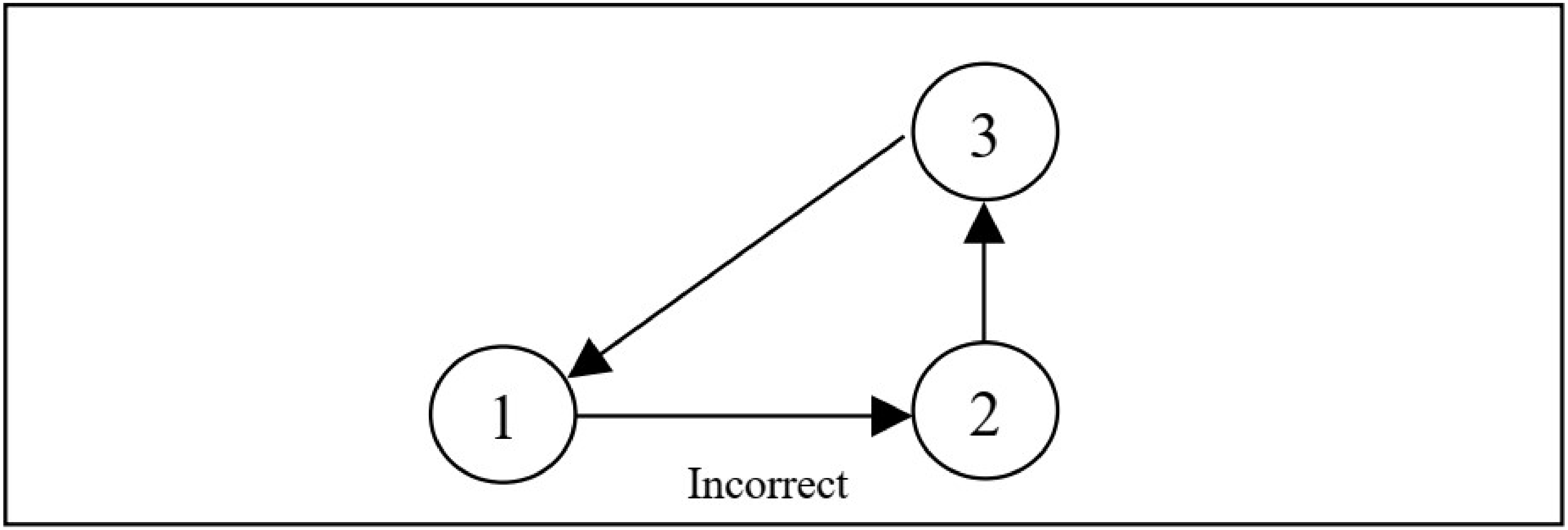
12.3 Errors to be Avoided in Constructing a Network (continued…)

b. Looping error should not be formed in a network, as *it represents performance of activities repeatedly in a cyclic manner* as shown below:



12.3 Errors to be Avoided in Constructing a Network (continued…)

c. The *direction of arrows* should flow *from left to right avoiding mixing of direction* as shown below:



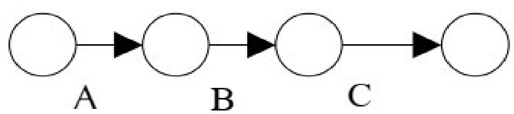
12.4 Rules in Constructing a Network

- An *activity* should *not be represented more than once in a network*.
- The *event numbered 1* is the *start event* and the *event with the highest number* is the *end event*.
- There should *not be any duplication of event numbers in a network*.
- *Dummy activities* must be *used only if it is necessary to reduce the complexity* of a network.
- A network should have *one start event* and *one end event*.

12.4 Rules in Constructing a Network (continued…)

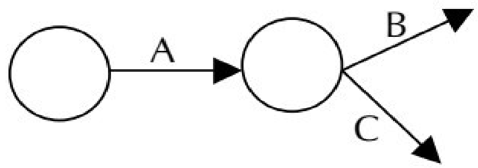
Some conventions of network diagram are shown below:

(a)



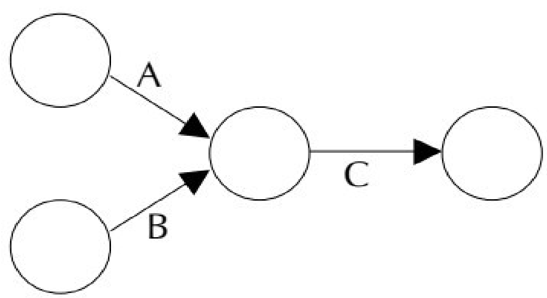
Activity B can be performed only after completing activity A, and activity C can be performed only after completing activity B.

(b)



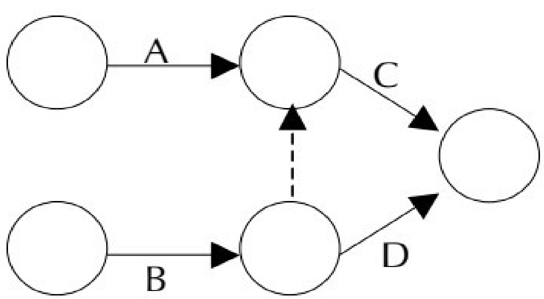
Activities B and C can start simultaneously only after completing A.

(c)



Activities A and B must be completed before start of activity C.

(d)



Activity C must start only after completing activities A and B. But activity D can start after completion of activity B.

12.5 Procedure For Numbering the Events using **Fulkerson's** Rule

Step 1: Number the start or initial event as 1.

Step 2: From event 1, strike off all outgoing event(s). Number that event as 2.

Step 3: Repeat step 2 for event 2, event 3 and till the end event. The end event must have the highest number.

12.5 Procedure For Numbering the Events using **Fulkerson's** Rule (continued…)

Example 1: Draw a network for a house construction project. The sequence of activities with their predecessors are given in Table 1 below: The network diagram is shown in next slide.

Name of the activity	Starting and finishing event	Description of activity	Predecessor	Time duration (days)
A	(1,2)	Prepare the house plan	--	4
B	(2,3)	Construct the house	A	58
C	(3,4)	Fix the door / windows	B	2
D	(3,5)	Wiring the house	B	2
E	(4,6)	Paint the house	C	1
F	(5,6)	Polish the doors / windows	D	1

Table 1

12.5 Procedure For Numbering the Events using **Fulkerson's** Rule (continued…)

Solution:

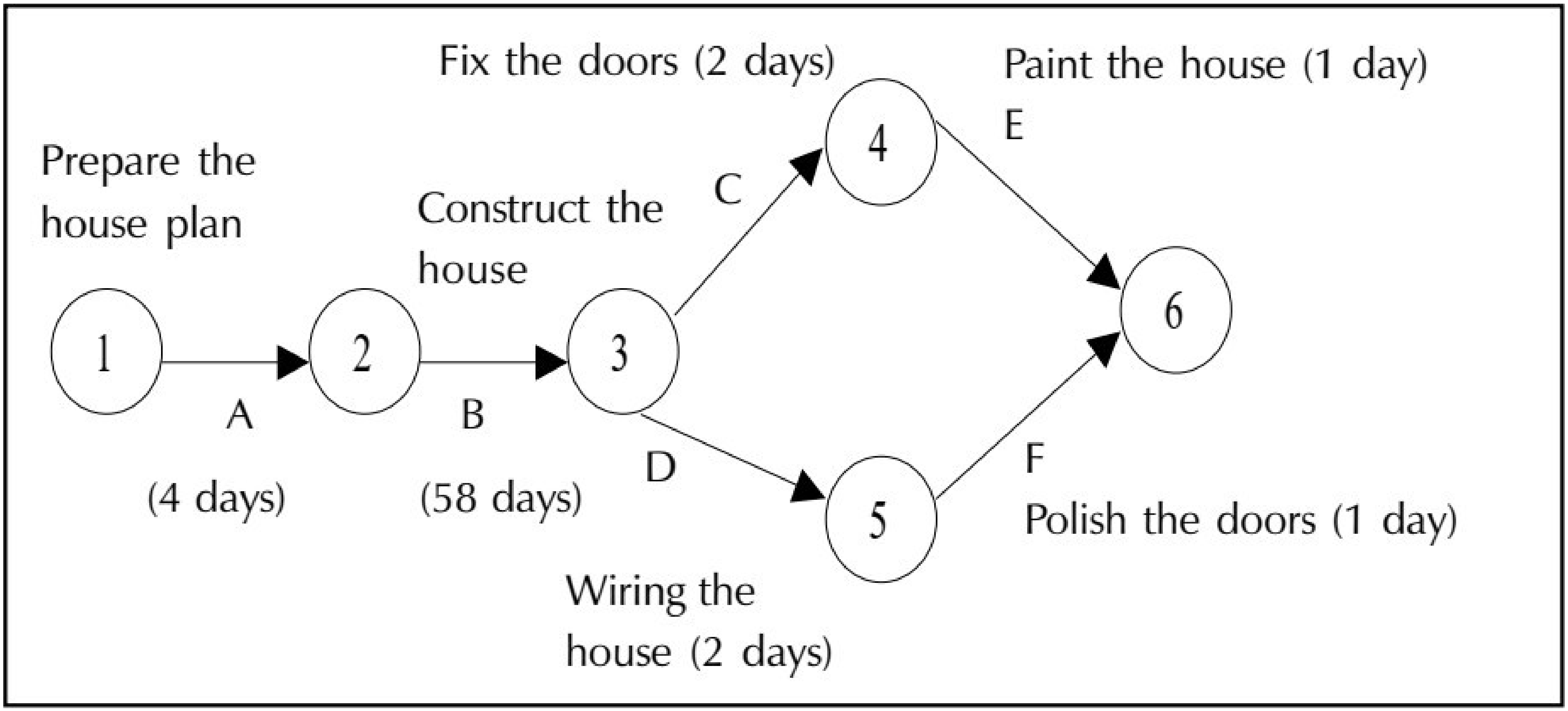


Figure 1

12.5 Procedure For Numbering the Events using **Fulkerson's** Rule (continued…)

Solution (continued):

The network diagram in Figure 1 shows the procedure relationship between the activities. Activity A (preparation of house plan), has a start event 1 as well as an ending event 2. Activity B (Construction of house) begins at event 2 and ends at event 3. The activity B cannot start until activity A has been completed. Activities C and D cannot begin until activity B has been completed, but they can be performed simultaneously. Similarly, activities E and F can start only after completion of activities C and D respectively. Both activities E and F finish at the end of event 6.

12.5 Procedure For Numbering the Events using **Fulkerson's** Rule (continued…)

Example 2: Consider the project given in table 2 and construct a network diagram

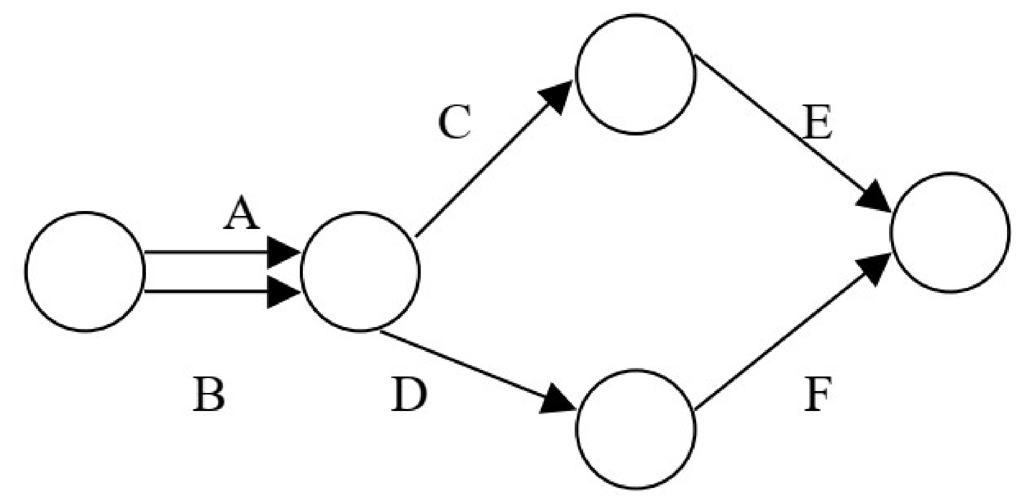
Activity	Description	Predecessor
A	Purchase of Land	-
B	Preparation of building plan	-
C	Level or clean the land	A
D	Register and get approval	A, B
E	Construct the building	C
F	Paint the building	D

Table 2

12.5 Procedure For Numbering the Events using **Fulkerson's** Rule (continued…)

Solution:

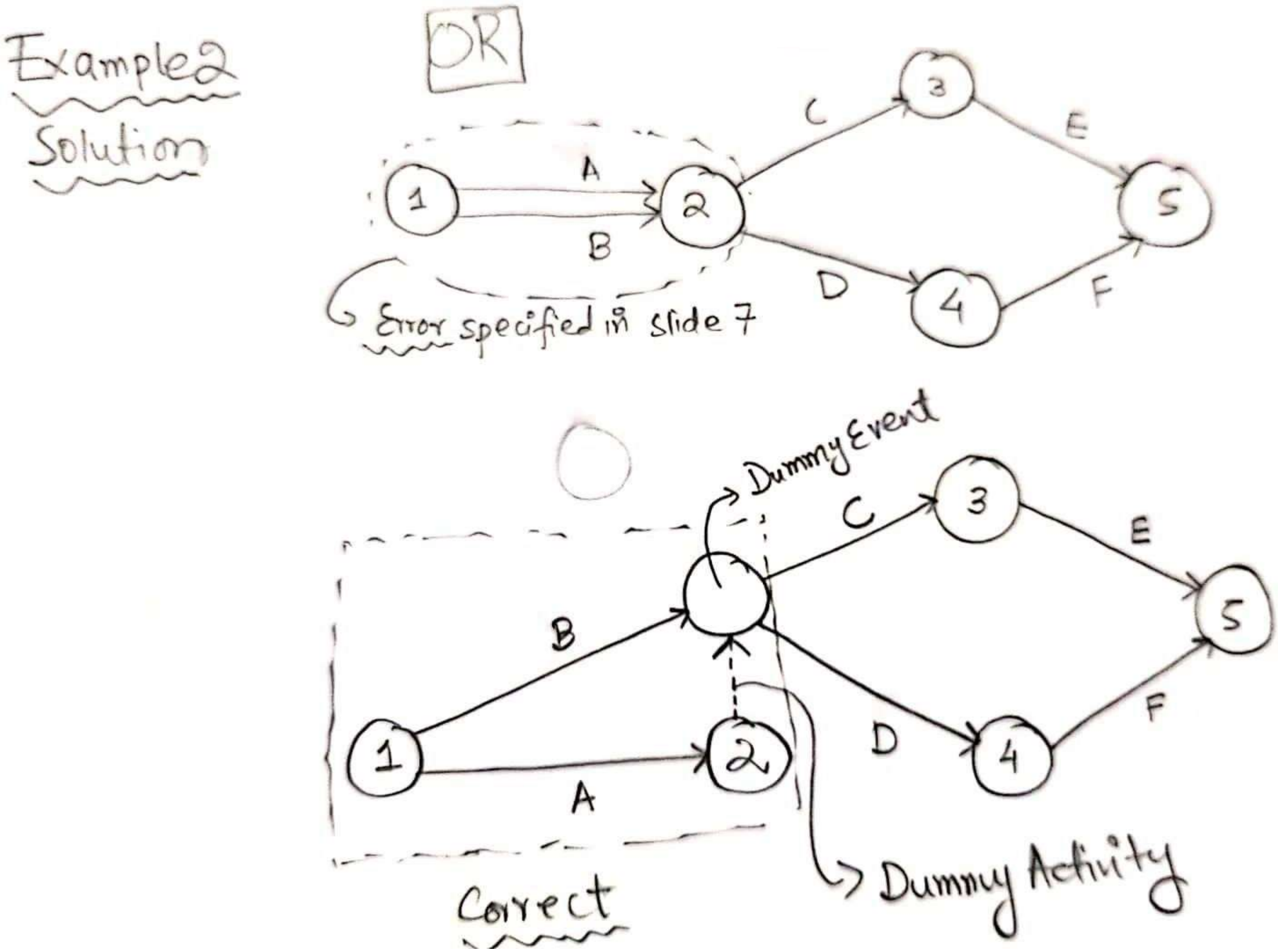
The activities C and D have a common predecessor A. The network representation shown in *figure (a) violates the rule that no two activities can begin and end at the same events*. It appears as if activity B is a predecessor of activity C, which is not the case. *To construct the network in a logical order, it is necessary to introduce a dummy activity*.



(a)

12.5 Procedure For Numbering the Events using **Fulkerson's** Rule (continued…)

Solution:



12.5 Procedure For Numbering the Events using **Fulkerson's** Rule (continued…)

Home Work

Example 3: Construct a network for a project whose activities and their predecessor relationship are shown in table 3

Activity	A	B	C	D	E	F	G	H	I	J	K
Predecessor	-	-	-	A	B	B	C	D	E	H, I	F, G

Table 3

Example 4: Draw a network diagram for a project given in table 4.

Activity	A	B	C	D	E	F	G	H	I	J	K	L
Immediate Predecessor	-	A	B	A	D	C, E	D	D	H	H	F, H	G, J

Table 4

12.6 Difference between **PERT** and **CPM**

- The main difference between these two techniques *lies in the way activity time durations are addressed*. The *accuracy of an activity's time estimate* usually depends on *the information available from previous projects*. If an activity has been performed before, *its duration can be more accurately predicted/estimated*. However, *activities of a new work*, which are difficult to measure or are dependent on other uncertain variables, *may have a wide range of possible time durations*.
- ***CPM*** is best utilized for *repetitive* and *non-complex* projects where time estimates can be made with some measure of certainty, e.g., a construction project. ***PERT*** on the other hand, is useful for *non-repetitive* and *complex* projects in which activity time estimates may vary over a range of possibilities, e.g., a research project.

12.6 Difference between **PERT** and **CPM** (continued…)

- ***PERT technique*** has been developed *to apply a statistical treatment to the possible range of activity time durations*. The range of activity times is expressed in terms of three time estimates which will be discussed in section 12.9 .
- ***CPM*** has been developed to take care of the *time-cost trade-off dilemma* often presented to project managers. This technique enables the project manager *to understand the effect of various project time cycles on direct and indirect cost*. Obviously, compressing the project duration will reduce indirect costs, but may increase direct costs.

12.7 Critical Path Analysis

- ***Project Duration and Critical Path:*** Once the arrow/network diagram has been developed and the duration of various activities is estimated. It is possible to calculate project duration. Further analysis of the network helps in evaluating information about *critical paths*, *critical activities*, *float*, *slack* *etc.* This information is useful for efficient management of projects.
- ***The Critical Path:*** A path in a network diagram is an unbroken sequence of activities directed from the origin of event (start event) to the terminal event (end event). *The longest (a path along which it takes the longest to traverse) continuous chain of activities through the network, starting from the first event to the last event is called the critical path.*
- *The activities lying on the critical path are called critical activities*, which govern the completion of the project. Any delay in their execution would result in delay in overall completion of the project.

12.7 Critical Path Analysis (continued…)

Scheduling of Activities: Earliest Time (T_E) and Latest Time (T_L)

In order to identify the critical path among several possible paths in a network diagram, the calculation of *earliest expected completion time of events (T_E)* and the *latest allowable time or latest completion time of events (T_L)* is done.

Forward Pass Computations (to calculate Earliest, Time T_E)

Procedure

Step 1: Begin from the start event and move towards the end event.

Step 2: Put $T_E = 0$ for the start event.

Step 3: Go to the next event (i.e node 2) if there is an incoming activity for event 2, add calculate T_E of previous event (i.e event 1) and activity time. ***Note:*** If there are more than one incoming activities, calculate T_E for all incoming activities and take the maximum value. This value is the T_E for event 2.

Step 4: Repeat the same procedure from step 3 till the end event.

12.7 Critical Path Analysis (continued…)

Scheduling of Activities: Earliest Time and Latest Time

Backward Pass Computations (to calculate Latest Time T_L)

Procedure

***Step 1*:** Begin from end event and move towards the start event. Assume that the direction of arrows is reversed.

***Step 2*:** Latest Time T_L for the last event is the earliest time. T_E of the last event.

***Step 3*:** Go to the next event, if there is an incoming activity, subtract the value of T_L of previous event from the activity duration time. The arrived value is T_L for that event. If there are more than one incoming activities, take the minimum T_E value.

***Step 4*:** Repeat the same procedure from step 2 till the start event.

12.7 Critical Path Analysis (continued…)

Example 5: A project schedule has the following characteristics as shown in Table 5 below:

Activity	Name	Time	Activity	Name	Time (days)
1-2	A	4	5-6	G	4
1-3	B	1	5-7	H	8
2-4	C	1	6-8	I	1
3-4	D	1	7-8	J	2
3-5	E	6	8-10	K	5
4-9	F	5	9-10	L	7

Table 5

i. Construct the PERT network.

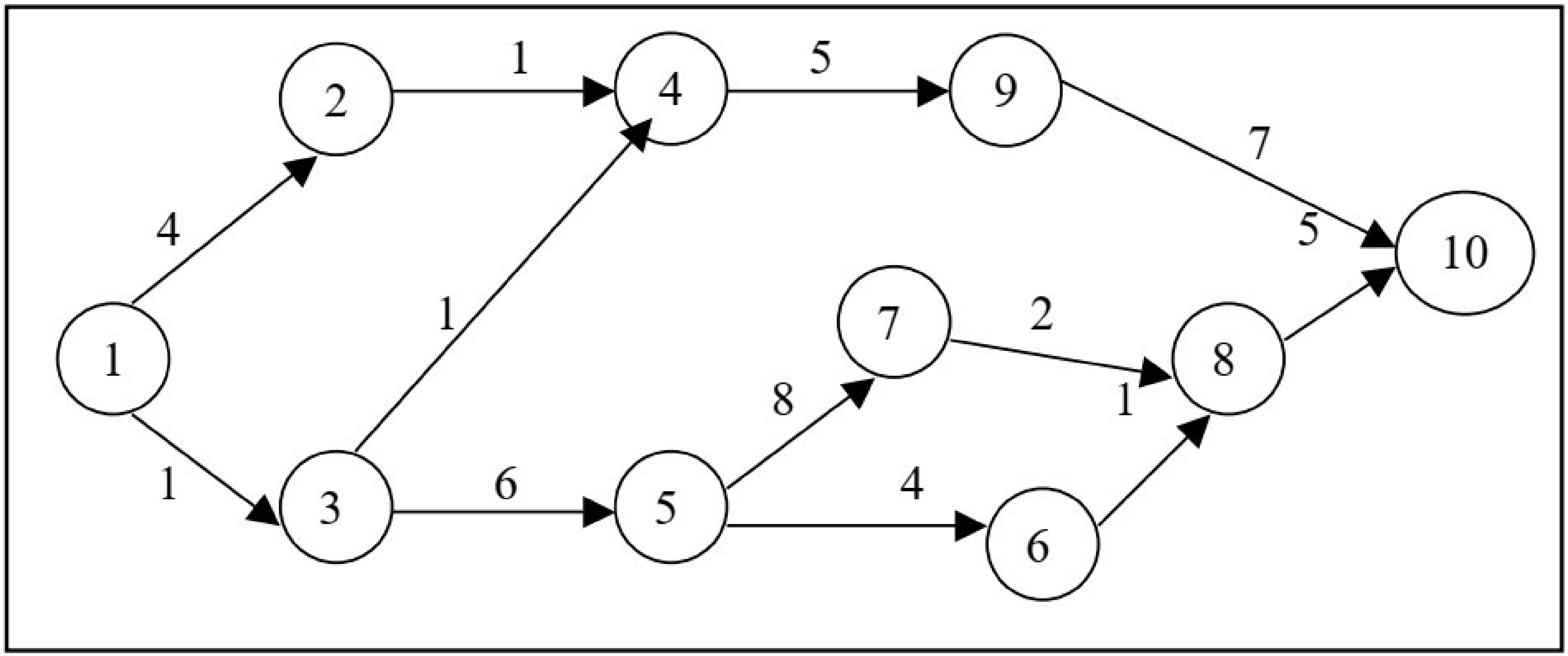
ii. Compute T_E and T_L for each activity.

iii. Find the critical path.

12.7 Critical Path Analysis (continued…)

Solution (continued):

(i): from the data given in table 5, the PERT chart is constructed as shown below:



PERT CHART / ACTIVITY NETWORK DIAGRAM

12.7 Critical Path Analysis (continued…)

Solution (continued):

(ii): To determine the critical path, compute the earliest, time T_E and latest time T_L for each of the activity of the project. The calculations of T_E and T_L are as follows:

To calculate T_E for all activities,

$$T_{E1} = 0$$

$$T_{E2} = T_{E1} + t_{1,2} = 0 + 4 = 4$$

$$T_{E3} = T_{E1} + t_{1,3} = 0 + 1 = 1$$

$$T_{E4} = \max (T_{E2} + t_{2,4} \text{ and } T_{E3} + t_{3,4})$$

$$= \max (4 + 1 \text{ and } 1 + 1) = \max (5, 2)$$

$$= 5 \text{ days}$$

$$T_{E5} = T_{E3} + t_{3,5} = 1 + 6 = 7$$

$$T_{E6} = T_{E5} + t_{5,6} = 7 + 4 = 11$$

$$T_{E7} = T_{E5} + t_{5,7} = 7 + 8 = 15$$

$$T_{E8} = \max (T_{E6} + t_{6,8} \text{ and } T_{E7} + t_{7,8})$$

$$= \max (11 + 1 \text{ and } 15 + 2) = \max (12, 17)$$

$$= 17 \text{ days}$$

12.7 Critical Path Analysis (continued…)

Solution (continued):

(ii):

$$T_{E9} = T_{E4} + t_{4,\,9} = 5 + 5 = 10$$

$$\begin{aligned} T_{E10} &= \max\,(T_{E9} + t_{9,\,10} \text{ and } T_{E8} + t_{8,\,10}) \\ &= \max\,(10 + 7 \text{ and } 17 + 5) = \max\,(17, 22) \\ &= 22 \text{ days} \end{aligned}$$

12.7 Critical Path Analysis (continued…)

Solution (continued):

(ii): To calculate T_L for all activities.

$T_{L10} = T_{E10} = 22$ days

$T_{L9} = T_{L10} - t_{9,10} = 22 - 7 = 15$ days

$T_{L8} = T_{L10} - t_{8,10} = 22 - 5 = 17$ days

$T_{L7} = T_{L8} - t_{7,8} = 17 - 2 = 15$ days.

$T_{L6} = T_{L8} - t_{6,8} = 17 - 1 = 16$ days

$T_{L5} = \min(T_{L7} - t_{5,7}$ and $T_{L6} - t_{5,6})$

$= \min(15 - 8$ and $16 - 4)$

$= \min(7$ and $11)$

$T_{L5} = 7$ days

12.7 Critical Path Analysis (continued…)

Solution (continued):

(ii):

$$T_{L4} = T_{L9} - t_{4,9} = 15 - 5 = 10 \text{ days}$$

$$T_{L3} = \min(T_{L4} - t_{3,4} \text{ and } T_{L5} - t_{3,5})$$

$$= \min(10 - 1 \text{ and } 7 - 6)$$

$$T_{L3} = 1 \text{ day}$$

$$T_{L2} = T_{L4} - t_{2,4} = 10 - 1 = 9 \text{ days}$$

$$T_{L1} = \min(T_{L2} - t_{1,2} \text{ and } T_{L3} - t_{1,3})$$

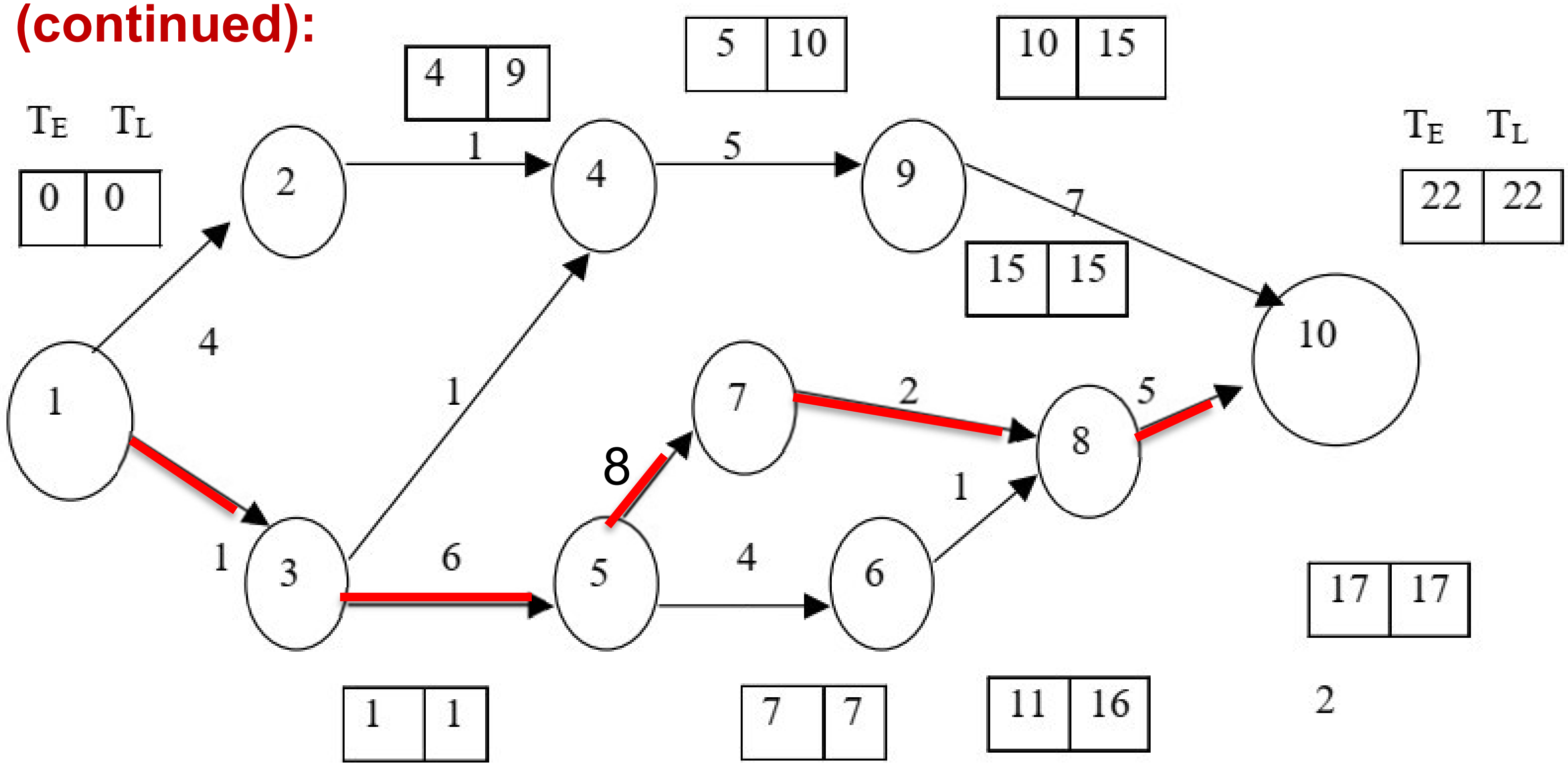
$$= \min(9 - 4 \text{ and } 1 - 1)$$

$$T_{L1} = 0$$

12.7 Critical Path Analysis (continued…)

Solution (continued):

(iii):



The critical path is 1-3-5-7-8-10 (as shown in red line above) with the project duration of 22 days.

12.8 Determination of Float and Slack Times

The activity, which does not lie on the critical path is called ***non-critical activity***. These non-critical activities may have some slack. The ***slack time*** is the *amount of time* by which the *start of an activity may be delayed without affecting the overall completion time of the project*.

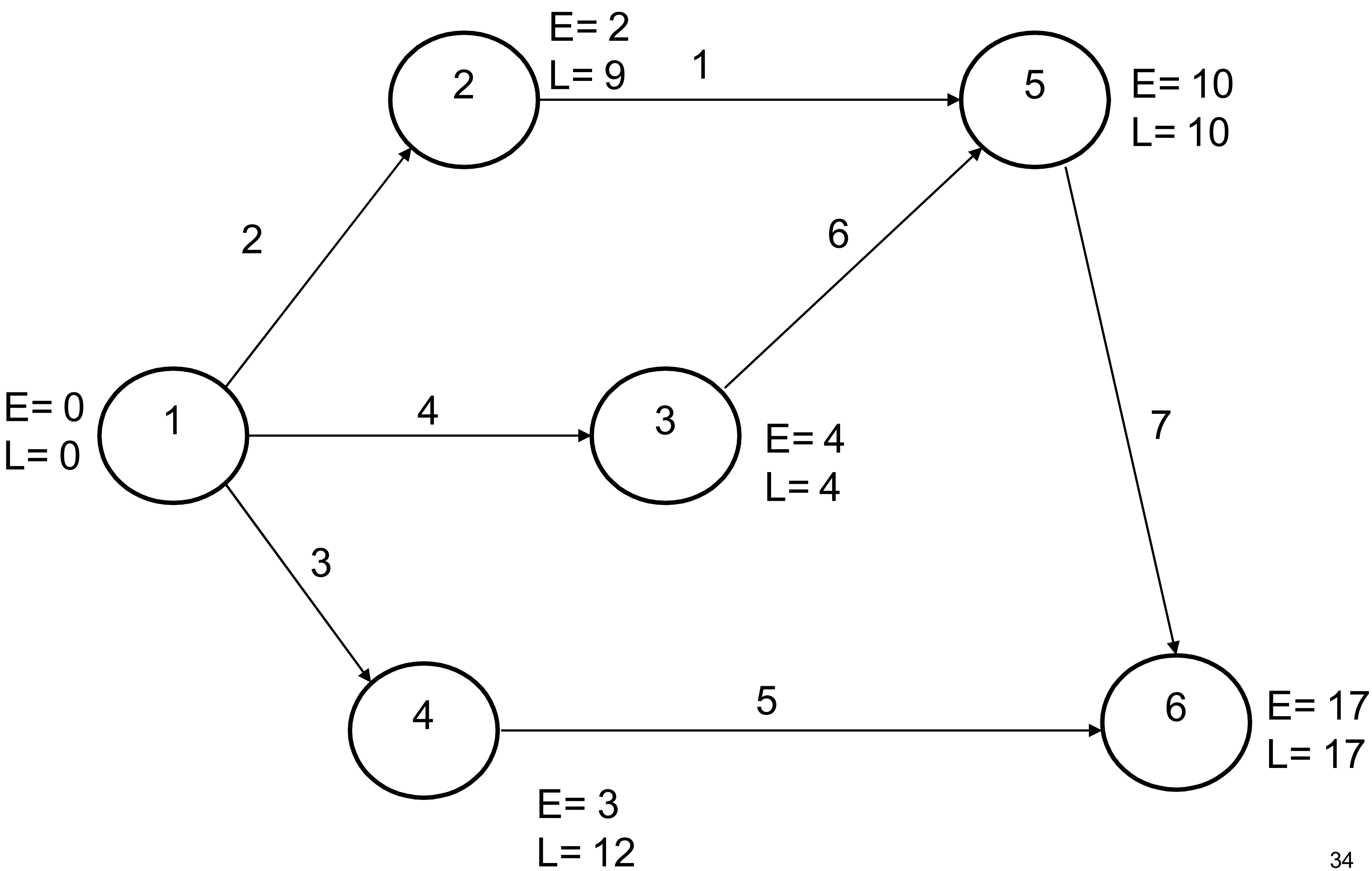
12.8 Determination of Float and Slack Times (continued…)

Head Event Slack (HES)

- The head event slack in a network is *the difference between the latest event time and earliest event time* at ***its head.*** It may be noted that HES should be calculated with the help of network after calculating Earliest Time (E) and Latest time (L).
- From the network in next slide HES of various activities are computed as:

Activity	HES = L - E at ***head node***
1-2	9 – 2 = 7
1-3	4 – 4 = 0
1-4	12 – 3 = 9
2-5	10 – 10 = 0
3-5	10 – 10 = 0
4-6	17 – 17 = 0
5-6	17 – 17 = 0

12.8 Determination of Float and Slack Times (continued…)



12.8 Determination of Float and Slack Times (continued…)

Tail Event Slack (TES)

- The tail event slack in a network is the *difference between the latest event time and earliest event time* at ***its tail***. It may be noted that TES should be calculated with the help of network after calculating Earliest Time (E) and Latest time (L).
- From the network in previous slide TES of various activities are computed as:

Activity	TES = L - E at ***tail node***
1-2	0 – 0 = 0
1-3	0 – 0 = 0
1-4	0 – 0 = 0
2-5	9 – 2 = 7
3-5	4 – 4 = 0
4-6	12 – 3 = 9
5-6	10 – 10 = 0

12.8 Determination of Float and Slack Times (continued…)

Earliest Finish Time (EFT)

The earliest finish time of an activity is equal to the *earliest start time of the activity* ***plus*** the duration of that activity. i.e., ***EFT = EST + Duration of that activity***.

Latest Start Time (LST)

The latest start time of an activity is equal to the latest finish time of the activity ***minus*** the duration of that activity . i.e., ***LST = LFT - Duration of that activity***.

Total Float (TF)

It refers to the *amount of free time* associated with *an activity* which *can be used before, during or after the performance of this activity*. TF is the *positive difference* between the earliest finish time (EFT) and the latest finish time (LFT), ***OR*** the positive difference between the earliest start time (EST) and the latest start time (LST) of an activity.

- ***TF = LFT – EFT***

  OR

- ***TF = LST - EST***

12.8 Determination of Float and Slack Times (continued…)

Free Float (FF)

Free float is that *portion of the total float* within which *an activity can be manipulated without affecting the float of subsequent activities*. It is computed for an activity by subtracting the HES from its TF. i.e., ***FF = TF - HES***

Independent Float (IF)

- IF is that *portion of the TF* within which *an activity can be delayed for start without affecting the float of the preceding activities*. It is computed for an activity by subtracting the tail TES from its TF. i.e., ***IF = TF - TES***. *Incase negative IF is obtained, it is taken as zero.*
- IF can also be computed using another formula i.e., ***IF = TF - FF***.

12.8 Determination of Float and Slack Times (continued…)

Example 6: A project has the following activities

Activity	Duration (Days)
1-2	2
1-3	4
1-4	3
2-5	1
3-5	6
4-6	5
5-6	7

12.8 Determination of Float and Slack Times (continued…)

Required

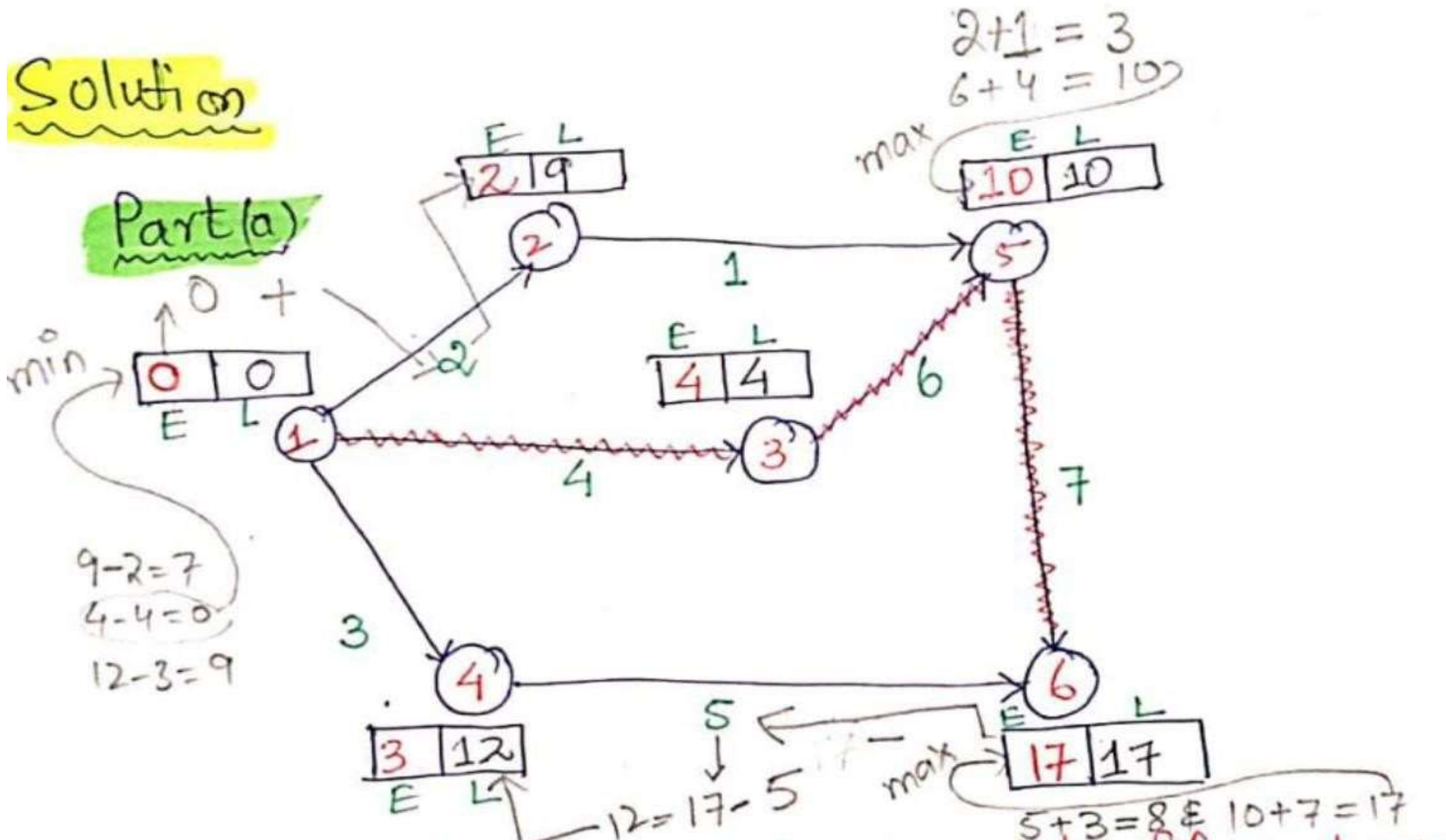
a) Draw the project network.

b) Find the critical path

c) Find earliest start time, latest start time, earliest finish time, and latest finish time.

d) Find total float, head even slack, free float, tail even slack and independent float

12.8 Determination of Float and Slack Times (continued…)



* We know for Forward Pass Computation, if we have more than one incoming activities, we select the maximum value.

12.8 Determination of Float and Slack Times (continued…)

* In Backward Pass Computation, if we have more than one incoming activities we select the minimum value.

Part(b)

Critical path is the longest continous path from node to node 6 shown in twisted line in network drawn in previous slide i.e.

red

$1 \rightarrow 3 \rightarrow 5 \rightarrow 6 = 4 + 6 + 7 = 17$ days

12.8 Determination of Float and Slack Times (continued…)

Activity	Duration (D)	ES /(EST) (Earliest start Time)	LF (Latest Finish) (LFT)	EF/(EFT) (Earliest Finish) (EST+D)	LS (Latest Start) (LFT-D)	TF (LF-EF)	HES (from Network)	FF (TF-HES)	TES (from Network)	IF (TF-FF)
1-2	2	0	9	0+2=2	9-2=7	9-2=7	9-2=7	7-7=0	0-0=0	7-0=7
1-3	4	0	4	0+4=4	4-4=0	4-4=0	4-4=0	0-0=0	0-0=0	0-0=0
1-4	3	0	12	0+3=3	12-3=9	12-3=9	12-3=9	9-9=0	0-0=0	9-0=9
2-5	1	2	10	2+1=3	10-1=9	10-3=7	10-10=0	7-0=7	9-2=7	7-7=0
3-5	6	4	10	4+6=10	10-6=4	10-10=0	10-10=0	0-0=0	4-4=0	0-0=0
4-6	5	3	17	3+5=8	17-5=12	17-8=9	17-17=0	9-0=9	12-3=9	9-9=0
5-6	7	10	17	10+7=17	17-7=10	17-17=0	17-17=0	0-0=0	10-10=0	0-0=0

Part (c) & (d) Solution

12.8 Determination of Float and Slack Times (continued…)

Home Work

Example 7: A project has the following activities.

Activity	Predecessors	Duration
A	-	7.8
B	-	20
C	-	33
D	A	18
E	A	20
F	B	9
G	C	9.8
H	D	8
I	E,F	4

12.8 Determination of Float and Slack Times (continued…)

Home Work

Example 7 (continued…):

Required

a. Draw the project network.
b. Find the critical path, Earliest Start Time, Earliest Finish Time, Latest Start Time, Latest Finish Time.
c. Find Total Float, Head Event Slack, Free Float, Tail Event Slack and Independent Float.

12.9 Project Evaluation and Review Technique (PERT)

- As the activity in PERT may be a novel one and that it wasn't performed in the past. PERT works on Beta (β) Probability.
- ***Time Estimates***

i. *Optimistic Time Estimate (t_o)*

The minimum time required for completion of the activity as per the pre-determined conditions.

ii. *Pessimistic Time Estimate (t_p)*

The maximum time that activity will take under worst conditions.

iii. *Most Likely Time Estimate (t_m)*

The time an activity will take if executed under normal conditions.

12.9 Project Evaluation and Review Technique (PERT) (continued…)

Basic Terms in PERT

i. Expected Time or Average time: $t_e = [t_o + 4t_m + t_p] / 6$

ii. Variance: $V = [(t_p - t_o) / 6]^2$

iii. Standard Deviation : $\sigma^2 = V_1 + V_2 + V_3$ (for critical activities)

iv. Probability of Completion of Project:

$$Z = (t - t_{cp}) / \sigma$$

where :

t = Pre-defined time

t_{cp}= Critical path time

12.9 Project Evaluation and Review Technique (PERT) (continued…)

Example 8 For the given activities determine:

a) Critical Path using PERT and also draw the network diagram

b) Calculate variance and Standard Deviation.

c) Calculate Probability of completing the project in 26 days.

12.9 Project Evaluation and Review Technique (PERT) (continued…)

Part (a)&(b) solution

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Activity	t_o	t_m	t_p	Duration= $t_e=\frac{t_o+t_p+4t_m}{6}$	EST or (ES)	EFT or (EF) (ES+D)	LST or (LS) (LF−D)	LFT or (LF)	Total Float (TF) (LF−EF)	Variance $V=\left(\frac{t_p-t_o}{6}\right)^2$
$\boxed{1-2}$	6	9	12	$=6+12+4(9)/6$ $=9$	0	9	0	9	0	$V=\left(\frac{12-6}{6}\right)^2$ $\boxed{V=1}$
1−3	3	4	11	5	0	5	4	9	4	1.778
$\boxed{2-4}$	2	5	14	6	9	15	9	15	0	$\boxed{4}$
3−4	4	6	8	6	5	11	9	15	4	0.444
3−5	1	1.5	5	2	5	7	20	22	15	0.444
2−6	5	6	7	6	9	15	18	24	9	0.111
$\boxed{4-6}$	7	8	15	9	15	24	15	24	0	$\boxed{1.778}$
5−6	1	2	3	2	7	9	22	24	15	0.111

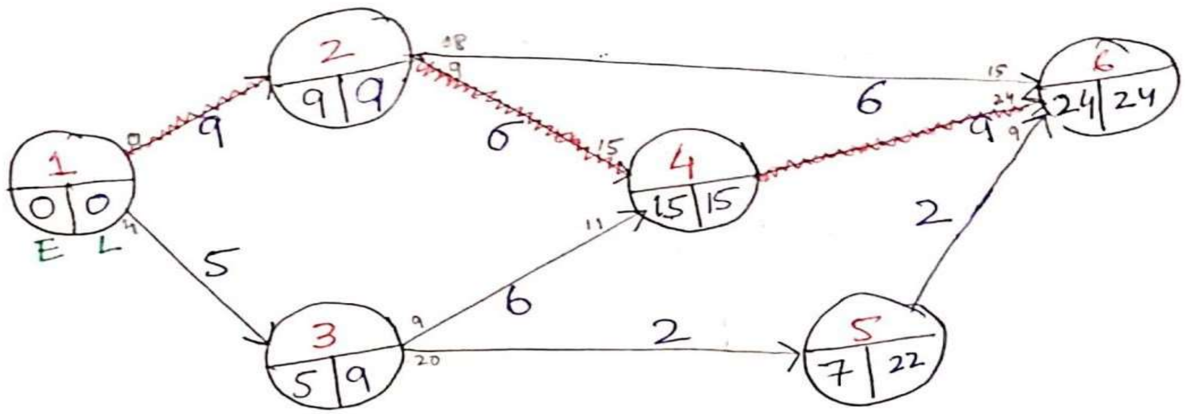
For Standard Deviation (σ)

→ It is square root of variance of only critical Activities and adding these variances as well.

$\sigma = \sqrt{1+4+1.778} = 2.6034$

12.9 Project Evaluation and Review Technique (PERT) (continued…)

Network Diagram



Critical Activities path

$\rightarrow$ 1 — 2 — 4 — 6 = 9 + 6 + 9 = 24 Days.

$\rightarrow \boxed{t_{cp} = 24 \text{ Days}}$

part(c) solution

t = pre-defined time = 26 days

t_{cp} = 24 Days.

$$Z = \frac{t - t_{cp}}{\sigma} = \frac{26-24}{2.6034} = 0.7682$$

$$Z = 0.7682$$

12.9 Project Evaluation and Review Technique (PERT) (continued…)

& Probability distribution table. we see

$z = 0.76$ & $\Psi(0.76) = 0.7764$

$z = 0.77$ & $\Psi(0.77) = 0.7794$

Take average of both values.

$$\Psi(z) = \frac{\Psi(z_1) + \Psi(z_2)}{2}$$

$$= \frac{0.7764 + 0.7794}{2}$$

$$\Psi(z) = 0.7779$$

$$\boxed{\Psi(z) = 77.79\%}$$

Probability distribution table is given at the end of these slides

12.9 Project Evaluation and Review Technique (PERT) (continued…)

Home Work

Example 9: A project has the following activities and characteristics:

Activity	*Optimistic Time Estimate(t_o)*	*Most Likely Time Estimate(t_m)*	*Pessimistic Time Estimate(t_p)*
1-2	2	5	8
1-3	4	10	16
1-4	1	7	13
2-5	5	8	11
3-5	2	8	14
4-6	6	9	12
5-6	4	7	10

12.9 Project Evaluation and Review Technique (PERT) (continued…)

Home Work

Example 9 (continued…):

Required

a. Draw the network diagram using PERT.
b. Calculate and identify the critical path obtained in part (a).
c. Calculate variance and standard deviation.
d. Calculate probability of completing of this project in 30 days.

12.9 Project Evaluation and Review Technique (PERT) (continued…)

APPENDIX - B

Proportion of total area under the normal curve from $-\infty$ to z.

where $z = \dfrac{x-\mu}{\sigma}$.

z	$\psi(z)$
0.00	0.5000
0.01	0.5040
0.02	0.5080
0.03	0.5120
0.04	0.5160
0.05	0.5199
0.06	0.5239
0.07	0.5279
0.08	0.5319
0.09	0.5359
0.10	0.5398
0.11	0.5438
0.12	0.5478
0.13	0.5517
0.14	0.5557
0.15	0.5596
0.16	0.5636
0.17	0.5675
0.18	0.5714
0.19	0.5753
0.20	0.5793
0.21	0.5832
0.22	0.5871
0.23	0.5910
0.24	0.5948
0.25	0.5987
0.26	0.6026
0.27	0.6064
0.28	0.6103
0.29	0.6141
0.30	0.6179
0.31	0.6217
0.32	0.6255
0.33	0.6293
0.34	0.6331
0.35	0.6368
0.36	0.6406
0.37	0.6443
0.38	0.6480
0.39	0.6517
0.40	0.6554
0.41	0.6591
0.42	0.6628
0.43	0.6664
0.44	0.6700
0.45	0.6736
0.46	0.6772
0.47	0.6808
0.48	0.6844
0.49	0.6879
0.50	0.6915
0.51	0.6950
0.52	0.6985
0.53	0.7019
0.54	0.7054
0.55	0.7088
0.56	0.7123
0.57	0.7157
0.58	0.7190
0.59	0.7224
0.60	0.7257
0.61	0.7291
0.62	0.7324
0.63	0.7357
0.64	0.7389
0.65	0.7422
0.66	0.7454
0.67	0.7486
0.68	0.7517
0.69	0.7549
0.70	0.7580
0.71	0.7611
0.72	0.7642
0.73	0.7673
0.74	0.7703
0.75	0.7734
0.76	0.7764
0.77	0.7794
0.78	0.7823
0.79	0.7852
0.80	0.7881
0.81	0.7910
0.82	0.7939
0.83	0.7967
0.84	0.7995
0.85	0.8023
0.86	0.8051
0.87	0.8078
0.88	0.8106
0.89	0.8133
0.90	0.8159
0.91	0.8186
0.92	0.8212
0.93	0.8238
0.94	0.8264
0.95	0.8289
0.96	0.8315
0.97	0.8340
0.98	0.8365
0.99	0.8389
1.00	0.8413
1.01	0.8438
1.02	0.8461
1.03	0.8485
1.04	0.8508
1.05	0.8531
1.06	0.8554
1.07	0.8577
1.08	0.8599
1.09	0.8621
1.10	0.8643
1.11	0.8665
1.12	0.8686
1.13	0.8708
1.14	0.8729
1.15	0.8749
1.16	0.8770
1.17	0.8790
1.18	0.8810
1.19	0.8830
1.20	0.8849
1.21	0.8869
1.22	0.8888
1.23	0.8907
1.24	0.8925
1.25	0.8944
1.26	0.8962
1.27	0.8980
1.28	0.8997
1.29	0.9015

12.9 Project Evaluation and Review Technique (PERT) (continued…)

APPENDIX - B

Proportion of total area under the normal curve from $-\infty$ to z.

where $z = \dfrac{x-\mu}{\sigma}$.

z	$\psi(z)$	z	$\psi(z)$	z	$\psi(z)$
0.00	0.5000	0.43	0.6664	0.85	0.8023
0.01	0.5040	0.44	0.6700	0.86	0.8051
0.02	0.5080	0.45	0.6736	0.87	0.8078
0.03	0.5120	0.46	0.6772	0.88	0.8106
0.04	0.5160	0.47	0.6808	0.89	0.8133
0.05	0.5199	0.48	0.6844	0.90	0.8159
0.06	0.5239	0.49	0.6879	0.91	0.8186
0.07	0.5279	0.50	0.6915	0.92	0.8212
0.08	0.5319	0.51	0.6950	0.93	0.8238
0.09	0.5359	0.52	0.6985	0.94	0.8264
0.10	0.5398	0.53	0.7019	0.95	0.8289
0.11	0.5438	0.54	0.7054	0.96	0.8315
0.12	0.5478	0.55	0.7088	0.97	0.8340
0.13	0.5517	0.56	0.7123	0.98	0.8365
0.14	0.5557	0.57	0.7157	0.99	0.8389
0.15	0.5596	0.58	0.7190	1.00	0.8413
0.16	0.5636	0.59	0.7224	1.01	0.8438
0.17	0.5675	0.60	0.7257	1.02	0.8461
0.18	0.5714	0.61	0.7291	1.03	0.8485
0.19	0.5753	0.62	0.7324	1.04	0.8508
0.20	0.5793	0.63	0.7357	1.05	0.8531
0.21	0.5832	0.64	0.7389	1.06	0.8554
0.22	0.5871	0.65	0.7422	1.07	0.8577
0.23	0.5910	0.66	0.7454	1.08	0.8599
0.24	0.5948	0.67	0.7486	1.09	0.8621
0.25	0.5987	0.68	0.7517	1.10	0.8643
0.26	0.6026	0.69	0.7549	1.11	0.8665
0.27	0.6064	0.70	0.7580	1.12	0.8686
0.28	0.6103	0.71	0.7611	1.13	0.8708
0.29	0.6141	0.72	0.7642	1.14	0.8729
0.30	0.6179	0.73	0.7673	1.15	0.8749
0.31	0.6217	0.74	0.7703	1.16	0.8770
0.32	0.6255	0.75	0.7734	1.17	0.8790
0.33	0.6293	0.76	0.7764	1.18	0.8810
0.34	0.6331	$\boxed{0.77}$	0.7794	1.19	0.8830
0.35	0.6368	0.78	0.7823	1.20	0.8849
0.36	0.6406	0.79	0.7852	1.21	0.8869
0.37	0.6443	0.80	0.7881	1.22	0.8888
0.38	0.6480	0.81	0.7910	1.23	0.8907
0.39	0.6517	0.82	0.7939	1.24	0.8925
0.40	0.6554	0.83	0.7967	1.25	0.8944
0.41	0.6591	0.84	0.7995	1.26	0.8962
0.42	0.6628			1.27	0.8980
				1.28	0.8997
				1.29	0.9015

12.9 Project Evaluation and Review Technique (PERT) (continued…)

APPENDIX - B

Proportion of total area under the normal curve from $-\infty$ to z.

where $z = \dfrac{x - \mu}{\sigma}$.

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0.04	0.5160	0.47	0.6808	0.89	0.8133
0.05	0.5199	0.48	0.6844	0.90	0.8159
0.06	0.5239	0.49	0.6879	0.91	0.8186
0.07	0.5279	0.50	0.6915	0.92	0.8212
0.08	0.5319	0.51	0.6950	0.93	0.8238
0.09	0.5359	0.52	0.6985	0.94	0.8264
0.10	0.5398	0.53	0.7019	0.95	0.8289
0.11	0.5438	0.54	0.7054	0.96	0.8315
0.12	0.5478	0.55	0.7088	0.97	0.8340
0.13	0.5517	0.56	0.7123	0.98	0.8365
0.14	0.5557	0.57	0.7157	0.99	0.8389
0.15	0.5596	0.58	0.7190	1.00	0.8413
0.16	0.5636	0.59	0.7224	1.01	0.8438
0.17	0.5675	0.60	0.7257	1.02	0.8461
0.18	0.5714	0.61	0.7291	1.03	0.8485
0.19	0.5753	0.62	0.7324	1.04	0.8508
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0.21	0.5832	0.64	0.7389	1.06	0.8554
0.22	0.5871	0.65	0.7422	1.07	0.8577
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0.28	0.6103	0.71	0.7611	1.13	0.8708
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0.34	0.6331	0.77	0.7794	1.19	0.8830
0.35	0.6368	0.78	0.7823	1.20	0.8849
0.36	0.6406	0.79	0.7852	1.21	0.8869
0.37	0.6443	0.80	0.7881	1.22	0.8888
0.38	0.6480	0.81	0.7910	1.23	0.8907
0.39	0.6517	0.82	0.7939	1.24	0.8925
0.40	0.6554	0.83	0.7967	1.25	0.8944
0.41	0.6591	0.84	0.7995	1.26	0.8962
0.42	0.6628			1.27	0.8980
				1.28	0.8997
				1.29	0.9015

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